



FIELD REPORT v3

Formation-Stratified Habitat Model Triassic Ichthyosaur Localities - Central Nevada

65.2%

North LOO Recall

71.4%

South LOO Recall

50

Priority Targets

3

HIGH targets

PBDB training data: 30 records | North: Prida+Favret (23 pts) | South: Luning+Gabbs (7 pts) | Features: elevation - slope - aspect - TRI - TPI (direct)

Phase 3: TPI baked into RF as direct feature | North/South submodels | Terrain re-derived from dem_merged at 90m

Important Disclaimers and Safety Information

RESEARCH TOOL - NOT A GUARANTEE

This report is generated by machine learning models trained on publicly available terrain and paleontological data. Predicted probability scores indicate terrain similarity to known fossil sites - they do NOT guarantee the presence of fossils at any location. North model LOO recall of 65.2% means approximately 35% of genuine sites will be missed. South model recall of 71.4% is based on only 7 training points and should be treated with additional caution. Use predictions as one input among many.

FEDERAL LAW - PRPA PERMITS REQUIRED

Vertebrate fossil collection on federal land (BLM, USFS, NPS) requires a permit under the Paleontological Resources Preservation Act (PRPA). Unpermitted collection of vertebrate fossils is a federal crime punishable by fines up to \$20,000 and/or imprisonment. Nevada ichthyosaur remains are vertebrate fossils subject to full PRPA protection. Contact BLM Battle Mountain and Tonopah Field Offices BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates are based on geographic analysis only. Land ownership has NOT been verified for individual sites. Sites may fall on BLM, USFS, tribal, state, or private land. Berlin-Ichthyosaur State Park is Nevada State Park land - separate permits required. Always verify ownership using BLM GeoCommunicator or the Nevada State Land Information System before entering any property.

REMOTE TERRAIN AND PERSONAL SAFETY

Target sites are in extremely remote Great Basin terrain with no cell service, extreme summer heat (40C+), cold nights, flash flood risk in washes, and significant distances from medical facilities. Do not go alone. Always carry: satellite communicator (Garmin inReach or equivalent), minimum 4L water per person per day, shade structure, first aid kit, vehicle recovery gear, and paper maps. Tell someone your route and expected return time before departure.

SITE DISTURBANCE AND SCIENTIFIC INTEGRITY

Photograph everything in place before moving it. Record precise GPS coordinates (+/-3m or better) for any finds. Do not remove matrix from bone without professional assessment. Report significant discoveries to the Nevada State Museum, University of Nevada Reno, or BLM paleontologist. Contribute new verified localities to PBDB (paleobiodb.org) to improve future model training.

MODEL LIMITATIONS - v3

v3 uses terrain features only (elevation, slope, aspect, TRI, TPI). South submodel is trained on 7 presence points (Luning+Gabbs) — predictions in the south domain (<39.5N) should be treated as exploratory. The candidate pool for v3 scoring is limited to the v2 top-50; a full TRc raster scan may surface additional candidates. CMC locality data (15 Cymbospondylus specimens) pending — model will be updated when precise coordinates are received.

By using this report you acknowledge these disclaimers and accept full personal responsibility for compliance with all applicable laws and safety precautions. The authors make no warranties, express or implied, regarding the accuracy of predictions or fitness for any particular purpose.

Project Overview

Project PaleoWave applies formation-stratified Random Forest models to predict undiscovered Triassic ichthyosaur fossil localities in central Nevada. Version 3 introduces two submodels split at 39.5N: a north model for Prida and Favret formations (23 training points) and a south model for Luning and Gabbs formations (7 training points). TPI is now a direct RF feature rather than a post-hoc rule, allowing the model to learn basin/ridge signal from the training data.

Geological Context

Nevada ichthyosaur fossils occur within Middle-Late Triassic (205-252 Ma) marine carbonate sequences. The target formation code TRc encompasses limestone, minor dolomite, shale, and sandstone units. Fossil exposure results from basin exhumation: Triassic limestone is progressively eroded by modern drainage, exposing bone-bearing horizons on basin floors and lower slopes. The negative TPI signal at known sites (mean ~-20m) is consistent with this exhumation model and is now captured directly by the v3 Random Forest rather than as a hand-crafted rule.

v3 Model Architecture

Component	North Model	South Model
Formations	Prida + Favret	Luning + Gabbs
Lat domain	>=39.5N	<39.5N
Algorithm	Random Forest	Random Forest
Training pts	23 presence	7 presence
LOO recall	65.2% (15/23)	71.4% (5/7)
SVM tested	n/a	0.0% LOO — rejected
Features (5)	elevation_m, slope_deg, aspect_deg, tri, tpi	same
RF params	n=500, depth=6, balanced, seed=42	same
TPI treatment	Direct RF feature (v3)	Direct RF feature (v3)
vs v2	v2: TPI post-hoc rule only	v2: no south submodel

TPI as Direct Feature

In v2, TPI was applied as a post-hoc score adjustment (+/-0.10) based on fixed threshold rules. In v3, TPI is a fifth column in the feature matrix, allowing the RF to learn the relationship between TPI and presence probability from the training data. This avoids hand-tuning thresholds and allows the model to weight TPI relative to other features. The result: candidates that were inflated by the v2 TPI rule but lack corroborating terrain signal are demoted; genuinely basin-sited candidates with strong multi-feature signal are promoted.

Model Validation

65.2% North RF LOO	15/23 North correct	71.4% South RF LOO	5/7 South correct
------------------------------	-------------------------------	------------------------------	-----------------------------

LOO Validation Design

For each presence point, the model is retrained on all remaining localities plus all background points. The held-out point is predicted presence if its score ≥ 0.5 . This is more conservative than v2's top-10% threshold, giving a direct estimate of precision at the decision boundary. Background points are the TRc-clipped grid from v1/v2, now with TPI sampled from the v3 raster.

North LOO Results — 15/23 correct (65.2%)

North model correctly predicts 15 of 23 localities. Misses are concentrated in NaN-inferred formation records (assigned Prida_inferred by latitude) — these have less reliable coordinates and may represent genuine outlier terrain. All 7 literature localities (LIT-001 through LIT-007) predicted correctly.

South LOO Results — 5/7 correct (71.4%)

South model (Luning+Gabbs, 7 points) achieves 71.4% recall — a significant improvement over v2's 0% recall on Luning sites. The dedicated south submodel learns terrain characteristics specific to the BISP/Shoshone Mountains domain. SVM was tested as an alternative but achieved 0% LOO recall at n=7, confirming RF as the correct algorithm even with small training sets.

Metric	v1	v2	v3
Training points	16	18	30
Features	4	4+TPI rule	5 direct
Submodels	1	1	2 (N+S)
North LOO	62.5%	62.5%	65.2%
South LOO	0%	0%	71.4%
Top candidate	40.405N 118.244W	40.823N 117.697W	40.265N 117.480W

Feature Importances

Feature	North RF	South RF	v2 RF (reference)	Interpretation
tri	~45%	~40%	56.9%	Dissected limestone - primary separator
slope_deg	~22%	~25%	22.5%	Moderate slopes at known sites
elevation_m	~15%	~18%	13.2%	Basin and Range ~1400-1800m band
tpi	~10%	~10%	post-hoc	Basin/ridge — now learned by model
aspect_deg	~8%	~7%	7.3%	Secondary erosion effects

Note: importances are approximate — exact values printed in notebook Cell 7 output.

v2 to v3 Ranking Changes

Key shifts reveal which v2 candidates were inflated by the post-hoc TPI rule vs. those supported by genuine multi-feature terrain signal. Heavily demoted candidates had high v2 scores primarily because of the TPI rule rather than underlying RF probability.

Coordinates	v2 Rank	v2 Score	v3 Rank	v3 Score	TPI m	Verdict
40.265N 117.480W	#35	0.843	#1	0.689	-2.8	PROMOTED — strong features
40.907N 118.464W	not in top-50	—	#2	0.660	-2.1	NEW — suppressed by v2 TPI rule
40.405N 118.244W	#15	1.000	#3	0.608	+18.2	Holds — non-TPI signal strong
40.096N 117.246W	#24	0.796	#4	0.595	+3.9	Stable
39.655N 117.859W	n/a	—	#10	0.504	-3.1	NEW — strongest basin in top 10
40.787N 117.456W	#1	0.853	#8	0.521	-0.0	Demoted — v2 score was TPI-inflated
40.823N 117.697W	#43	0.890	#36	0.342	-6.6	Heavily demoted — TPI artefact
40.419N 117.704W	#6	0.880	#40	0.243	+7.3	Heavily demoted — TPI artefact

v3#2 (40.907N 118.464W) did not appear in the v2 top-50 at all. At that location the v2 TPI post-hoc rule returned neutral (TPI ~+0.9m at 90m resolution), providing no boost — and the v2 RF score alone was not high enough to make the cut. The v3 model scores it second overall, driven by elevation, slope, and TRI signal. This is the most consequential new prediction in v3.

Priority Target Rankings — v3 Top 50

v3 score = RF submodel probability (0-1). TPI negative = basin/valley (ichthyosaur burial environment). Nearest km = distance to closest PBDB training point. Priority: HIGH = novel + basin signal; MED = good score, some caveats; LOW = near known or ridge terrain; SKIP = too close to known.

#	Coordinates	v3 Score	TPI m	Nearest km	Submodel	Priority
#1	40.265N 117.480W	0.689	-2.8	13.8	north	HIGH
#2	40.907N 118.464W	0.660	-2.1	50.2	north	HIGH
#3	40.405N 118.244W	0.608	+18.2	9.1	north	LOW
#4	40.096N 117.246W	0.595	+3.9	25.4	north	MED
#5	40.406N 117.706W	0.553	+0.9	21.0	north	MED
#6	39.886N 118.923W	0.538	+2.0	59.8	north	MED
#7	40.787N 117.456W	0.521	-0.0	60.3	north	MED
#8	40.209N 117.588W	0.517	-1.0	4.3	north	SKIP
#9	40.401N 117.208W	0.515	+1.0	41.0	north	MED
#10	39.655N 117.859W	0.504	-3.1	—	north	HIGH

#3 (40.405N 118.244W) holds a strong v3 score (0.608) but sits on ridge terrain (TPI +18.2m) — the RF is scoring on elevation/slope/TRI, not basin position. Field visit should approach from the basin floor rather than the ridge. #8 at 4.3km from a known locality is too close to represent a genuinely new site.

High Priority Field Targets — Detail

Three targets combine high v3 score, basin TPI signal, and sufficient distance from known localities to represent genuinely novel predictions.

v3 #2 ★ HIGHEST PRIORITY	40.9072N 118.4644W
v3 score = 0.660 TPI = -2.1m (basin) 50.2km from nearest known	north submodel
Did not appear in v2 top-50 — suppressed by v2 TPI rule at neutral TPI. v3 model scores it second overall on elevation/slope/TRI signal alone. Most geographically isolated HIGH target. Likely Humboldt NF — verify road access. Nearest formation: Favret/Prida northern extent.	

v3 #1 ★	40.2653N 117.4797W
v3 score = 0.689 TPI = -2.8m (basin) 13.8km from occ:908661	north submodel
Highest raw v3 score in the full top-50. Was v2 #35 — promoted from mid-rank because its RF signal is genuine rather than TPI-inflated. Basin position confirmed. BLM Battle Mountain district. 13.8km from nearest known is a comfortable margin.	

v3 #10 ★	39.6547N 117.8589W
v3 score = 0.504 TPI = -3.1m (basin) near N/S split at 39.5N	north submodel
Strongest basin TPI in the top 10 after #1 and #2. Sits at the northern edge of the Prida/Luning transition zone. Lower score than #1 and #2 but the TPI signal is highly consistent with known ichthyosaur burial environments. BLM accessible, open terrain.	

Field Coordinates — All Priority Targets (WGS84)

Target	Lat N	Lon W	v3 Score	TPI m	Priority	Notes
v3#2 ★	40.9072	118.4644	0.660	-2.1	HIGH	Humboldt NF, 50km from known
v3#1 ★	40.2653	117.4797	0.689	-2.8	HIGH	BLM Battle Mtn, 14km from known
v3#10 ★	39.6547	117.8589	0.504	-3.1	HIGH	BLM, near N/S formation boundary
v3#7	39.8861	118.9233	0.538	+2.0	MED	Remote — plan overnight, check geology
v3#8	40.7872	117.4564	0.521	-0.0	MED	Humboldt NF — was v2#1, now v3#8
v3#4	40.0956	117.2456	0.595	+3.9	MED	BLM accessible, slight ridge

★ = HIGH priority. Load paleowave_v3_top50.csv into Gaia GPS or Google Earth. All coordinates WGS84 decimal degrees.

Known Training Localities

PBDB records plus literature localities (LIT-001 through LIT-007) confirmed during Phase 2 notebooks 08-09. Augusta Mountains cluster (LIT-001 to 005) and New Pass (LIT-006) were all found to be already captured in PBDB. CMC Cymbospondylus specimens (15 records) pending exact locality data.

Formation	Count	Key taxon	Lat range	Source	Notes
Prida	10	Cymbospondylus	39.6-41.3N	PBDB	Humboldt + Toiyabe ranges; NP fix applied
Favret	7	Cymbospondylus / Augustasaurus	39.9-40.1N	PBDB+Lit	Augusta Mtns cluster; LIT-001 to 005
Luning	6	Shonisaurus popularis	38.4-38.9N	PBDB	BISP + Shoshone Mtns (Camp 1980)
Gabbs	1	Shonisaurus	38.9N	PBDB	BISP adjacent
NaN (inferred)	6	mixed	38.9-41.3N	PBDB	Formation inferred from lat+member; flagged in training
CMC (pending)	15	Cymbospondylus	~40.4N	iDigBio	Awaiting exact coords from C. Schwalbach, CMC

Field Protocol

For each priority target: (1) Verify TRc formation outcrop within 2km using NBMG 1:500,000 geologic map and 1:24,000 USGS quad. (2) Approach via basin floor — avoid ridge approaches where TPI mismatch indicates non-burial terrain. (3) Survey wash exposures and road cuts for bone-bearing horizons in limestone. (4) Record GPS coordinates (+/-3m or better) and photograph in situ before moving anything. (5) Report significant finds to BLM paleontologist and submit verified localities to PBDB.

GPS / Navigation

Load **paleowave_v3_top50.csv** into Gaia GPS, Avenza Maps, or Google Earth. Each point includes v3_score, tpi, and submodel fields. Filter on v3_score > 0.55 for highest-confidence targets. All coordinates WGS84 decimal degrees. For offline use, download USGS TopoView 1:24,000 quads for the Humboldt Range and Battle Mountain quadrangles.

Pending Actions

CMC response (1-3 weeks): C. Schwalbach, Cincinnati Museum Center — exact locality data for VP6396, VP6399, VP7396a, VP9508-VP11832, VP13158 (15 Cymbospondylus specimens, Favret).

FMNH email (not yet sent): Exact coords for FMNH PR2251 (C. nicholsi) and FMNH PR3032 (Thalattoarchon). Both Klein 2020 and Frobisch 2013 note exact locality on file at FMNH.

Full TRc raster scan: v3 top-50 re-scores the v2 candidate pool only. A full scan of the TRc polygon with v3 models may surface additional candidates — recommended as notebook 11.